Project: Design and Development of a Bidirectional DC-DC Power Shuffler for Partial Shading Mitigation in Photovoltaic Systems

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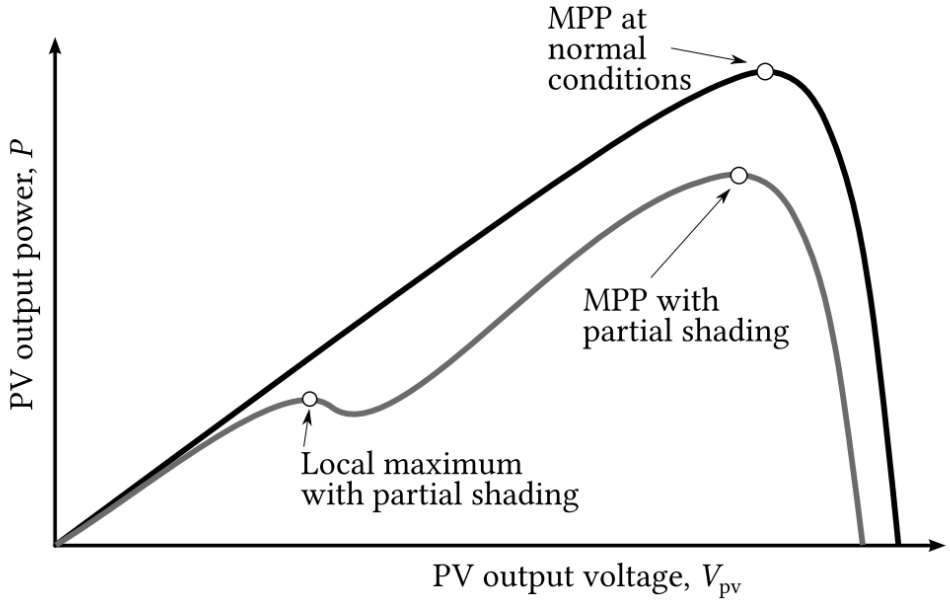
Version	Date	Author	Changes/Comments
1	17/4/2026	Son Nguyen	Initial Revision

General Objective

The objective of this project is to design and develop a robust and scalable hardware for the purpose of power optimisation from photovoltaic cells. The hardware is to take the form of a bidirectional DC-DC converter which is to be used in tandem with existing or new photovoltaic panels in multiple applications such as commercial and residential array, as well as portable applications such as for use in camping. The purpose of the hardware to ensure maximum power generation from PV modules in partial shading conditions such as shadows from trees shading a portion of the PV array such as in early morning or late afternoon conditions where the sun is close to the horizon. The project will explore critical challenges such as appropriate components selection, hardware topology, PCB design and integration with a wide range of PV modules. Additionally, this project will investigate the best practices on integrating the prototype hardware into existing PV modules, allowing for commercialisation of the solar optimisation hardware in applications where maximum efficiency is of utmost importance. The key research question of this project is: How can a scalable, non-isolated bidirectional DC-DC power shuffler be systematically designed, simulated, and validated to mitigate the effects of partial shading on photovoltaic string output, and what hardware topology and component selection strategies best balance efficiency, cost, and compatibility with commercially available PV modules?

Key Finding from the Literature - Impact on Your Project Design

Over the last decade, PV installations have grown rapidly on a global scale, with annual deployments increasing 15-fold from 40GW in 2014 to 602GW in 2024. From an Australian perspective, annual solar electricity generation revenue has grown from $100 Million AUD in 2015 to $2.1 Billion AUD in 2025 [1]. Despite these gains in scale and cell efficiency, the fundamental electrical topology of a PV array has remained largely unchanged, with cells connected in series within a panel, panels connected in series to form a string, and strings connected in parallel to form an array [2]. Whilst this topology is simple and well understood, it suffers from a fundamental limitation in that the output current of a series connection is dictated by the lowest performing module in the string. As a result, a single shaded cell, a soiled panel, or a mismatched module can disproportionately reduce the output of an entire string. On a modern 72-cell PV panel with conventional bypass diode layout, a single shaded cell can cause a one-third reduction in the panel's maximum output power [3].



The effect of partial shading also extends to Maximum Power Point Tracking (MPPT) performance. Under uniform irradiance, a PV string presents a single global maximum on its power-voltage curve, and low-complexity algorithms such as Perturb and Observe (P&O) are highly effective at tracking this maximum. However, under partial shading, the string's power-voltage curve presents multiple local maxima, and P&O algorithms are susceptible to locking onto a local maximum rather than the global maximum, leading to further reductions in extracted power [4]. Whilst more sophisticated MPPT algorithms have been proposed in the literature to address this limitation, these approaches add significant computational complexity and cost to the system without fundamentally solving the underlying current mismatch between modules at the hardware level.

Three primary approaches have emerged in the literature for mitigating partial shading at the system level, namely array reconfiguration (both dynamic and static), advanced control system algorithms, and power shuffler devices [5][6][7]. Whilst array reconfiguration is theoretically effective, it introduces substantial wiring complexity, additional switching hardware and installation cost which has limited its real-world adoption. Advanced MPPT algorithms address the symptom rather than the cause of the problem, and their effectiveness is ultimately bounded by the underlying current mismatch between modules [3]. Power shufflers, on the other hand, address the mismatch problem directly at the hardware level by redistributing current between modules so that

every panel in the string operates close to its individual maximum power point despite differences in irradiance [8].

The classification framework presented by Kasper et al. [9] has been particularly valuable in understanding the design space for panel-integrated DC-DC converters. The categorisation of converters into full-power and partial-power architectures, with further subdivision into series and parallel connected topologies, provides a structured basis for evaluating the trade-offs between different design choices. Full-power converters offer flexibility and simplicity but incur conversion losses on the entire panel output even under uniform irradiance. Partial-power converters offer an inherent efficiency advantage by processing only the mismatch current between modules, as the remainder of the power flows directly from panel to string output through a highly efficient bypass path[9][10]. Within the partial-power category, parallel-connected partial-power converters (P-PPCs) have been identified as the most promising architecture for this project as they operate on the principle of voltage equalisation between adjacent panels and can be realised using non-isolated bidirectional topologies [11].

Building on the foundational work by Walker [12], the P-PPC architecture has been selected as the basis for the power shuffler design in this project. The bidirectional buck-boost and Ćuk converter topologies have been identified as the leading candidates for the final design, as both are capable of bidirectional current shuffling between adjacent modules without the need for an isolation transformer [13]. The final topology selection will be made during the simulation phase of the project based on factors such as component stress, ripple current, efficiency under various operating conditions, and ease of integration with commercially available PV modules. The scalable and modular nature of the P-PPC architecture is also particularly well suited to the project's goal of producing a hardware platform which is compatible with a wide range of PV module ratings and deployment scenarios, from residential rooftop installations through to off-grid portable applications.

Stakeholders & Resources

Stakeholders:

Internal stakeholders

The internal stakeholders of this project are Son Nguyen and Jared Saunders, student engineers responsible for the power electronics and control systems streams respectively, and the Geoff Walker, the academic supervisor who provides technical guidance and approval at key design milestones. Given the interdependency between the two streams, both students are considered active stakeholders in each other's progress, particularly at interface points such as communication protocol selection and microcontroller integration. QUT's Power Engineering Research Division is also considered an internal stakeholder, as the facility and its equipment will be utilised for live PV string testing in Semester 2.

External Stakeholders

The broader academic research community working in the field of differential power processing and PV optimisation represents the primary external stakeholder group. Researchers investigating partial shading mitigation strategies, including power shuffler topologies and advanced MPPT algorithms may have an interest in the outcomes and findings of this project, particularly given the limited body of experimental hardware validation work in this space. Additionally, residential and commercial

solar users stand as indirect external stakeholders, as the hardware developed in this project is ultimately aimed at improving energy extraction efficiency real-world PV installations.

Resources:

Hardware

- Programmable DC power supplies and electronic loads for emulating PV source behaviour and MPPT demand during bench testing
- Oscilloscopes for switching waveform inspection and verification against simulation results
- IR imaging equipment for thermal characterisation under sustained load
- Multichannel data acquisition systems and power analysers for live PV string testing at QUT's Power Engineering Research Facility
- Commercially available components procured for PCB assembly, with fabrication outsourced to a third-party manufacturer

Software

- Qspice will be used for circuit simulation and hardware in the loop testing once integrated with control systems.
- Kicad is preferably used for PCB Design due to its free nature.

Funding

- University project allocation to cover the majority of component and PCB fabrication costs
- Personal student contribution anticipated for costs exceeding the university allocation
- Detailed budget to be confirmed with supervisor prior to the first PCB order

Data

- Data sheets on all electronic components of the power shuffler and MPPT
- Data sheets of PV modules to be used in field testing.
- Common irradiance and temperatures for PV operations
- Experimental data such efficiency, losses and V-A curves when testing components individually and part of a string

Ethics

- Electrical safety during live PV testing, which is addressed in the Risks, Requirements & Constraints section

Project Methodology

The project will begin with a market study of commercially available PV modules to establish the input specifications for the power shuffler. By sizing the hardware around the highest output modules currently on the market, the design will maintain broad compatibility with lower-rated

modules while remaining relevant against the industry trend of increasing panel power output over time.

Accompanying the power shuffler will be the development of a basic MPPT boost converter. Whilst the power shuffler remains the primary focus of this project, a typical PV string deployment pairs the array with a DC-DC boost converter to step up the string voltage to the desired bus level, and an MPPT controller is required to continuously track the string's maximum power point. Without this, the power shufflers would operate without visibility of the string's overall power extraction status, undermining the ability to properly evaluate their performance. The MPPT boost converter therefore represents a secondary but necessary deliverable and will be developed to a sufficient level of functionality to support system-level validation of the power shuffler rather than as a standalone research focus.

To gain an initial understanding of the power shufflers topologies mentioned in the literature review, circuit simulations are conducted to gain key insight into key performance metrics such as ripple current and output voltage. For this purpose, Qspice was chosen as the simulation tool due to its free price, open-source nature and C code integration which allows for greater modelling flexibility in comparison to other SPICE programs such as LTspice. To ensure accurate representation of PV modules output and their interaction with the power shufflers, a PV module will also be modelled in Qspice and used in circuit simulations. Once the behaviour of different shuffler topologies are simulated, a topology will be chosen for the final design based on factors such as component ratings, component stress and ease of integration with PV modules. Upon selection of a topology, accurate modelling of components such as switches will occur to determine losses and identify potential problems with the circuit. This phase is finished once the circuit design is validated and the schematics are ready for the first PCB layout.

The MPPT boost converter circuit simulation will be based on existing designs to reduce their development time. Given the technological maturity of MPPT boost converter topologies, development effort in this area yields comparatively less efficiency gain than equivalent effort directed toward the power shuffler. As such, the MPPT is scoped as a supporting deliverable and will be developed to a sufficient level of functionality to support system-level validation of the power shuffler.

Following simulation validation, the hardware is developed using KiCad across multiple PCB revisions. The first revision is designed specifically as a diagnostic platform, incorporating extensive test points at all critical nodes such including gate drive outputs and switching nodes. This revision will also have generous clearances for safe oscilloscope probing and uses basic headers connectors for control systems integration from a microcontroller such as the STM-32. This decision is made on the assumption that microcontrollers selection is unlikely to have occurred at this point in the project. The lessons learned from this initial revision informs the second revision, which aims to resolve key identified issues and add basic communication capability as well as microcontroller integration, allowing for operation in pairs on a dual PV module system. A third version will then be developed with more robust communication systems (e.g. CAN), allowing for master-slave capability for system integration with our custom MPPT.

The MPPT system will follow a similar design concept with version one acting as a test bed and designed to operate in isolation. Unlike our power shuffler, we aim to integrate a microcontroller into our version one of our MPPT as their operations and designs are well documented. Areas of improvement will then be identified following testing, leading to development of a follow design which allows for multi MPPT operations, such as in an extended string or multi-string set-ups. As

such, version two of our power shuffler is designed to work with version three of our power shuffler due to the prior two version lacking master-slave capability.

Upon receipt of the fabricated PCBs, individual shuffler will then undergo bench testing using programmable DC power supplies to emulate PV source behaviour and electronic load to simulate MPPT demands. This testing set-up allows for precise mapping of the power shuffler's efficiency across varying PV's output and the thermal management solution (heatsinks) via IR imaging across sustained load. Oscilloscope measurements are used to inspect for abnormal switching behaviours or excessive voltage/current ripple. The data obtained from these tests are used to verify switching losses and compare the power shuffler's behaviour against the circuit simulation in Qspice.

A similar testing method will be utilised for our MPPT controller where version one will be bench-tested using DC power supplies to simulate PV modules. If there are no critical issues identified from bench testing, the MPPT is then tested with real PV modules to gather additional data required for development of the second version.

Following the testing of individual units, multiple shufflers are integrated into a real PV string for live testing along with the updated (V2) MPPT with the aim of evaluating system level behaviour. Due to the high-risk factors arising from the connected PV string such as elevated voltage and high output power, testing will be performed at QUT Power Engineering Research Facility to better contain hazards. For testing in this stage, power analysers and multi-channel data acquisition systems will be utilised to collect concurrent data such as current flow (shuffle) between different panels. The primary goal of this phase is the assessment of total system efficiency, stability and the tracking of power voltage curve of the solar string. Final validation is achieved through an outdoor test campaign, where the system is exposed to dynamic irradiance and real-world module impedance. Data from both laboratory and field tests is logged and processed using MATLAB or Python, where statistical analysis and uncertainty calculations are applied to quantify the performance gain. This comprehensive testing suite ensures that the system is fully validated from a component level through to a complete, integrated power system.

Deliverables

Semester 1:

- Circuit Simulations
    - Basic circuit simulations of various power shuffler topologies when interfacing with two PV modules.
    - Detailed circuit simulations of chosen power shuffler topology when interfacing with two and four module system. Includes details such as realistic switching devices (MOSFETs) and losses
    - Basic and detailed circuit simulation of MPPT controlled boost converter based on existing devices
- Hardware Design and Testing
    - Initial PCB Design of Power Shuffler and MPPT for supervisor review
    - Final PCB Design Power Shuffler and MPPT sent for manufacturing
    - Data from bench testing of power shuffler version one
- Documentation
    - Literature review and project proposal
    - Interim project report
    - Presentation slides

Summer Break:

- Hardware Design and Testing
    - Updated design of power shuffler (version two) sent to supervisor for approval and manufacturer
    - Draft PCB design of MPPT version one sent to supervisor for review
    - Final design of MPPT version one sent for manufacturing

Semester 2:

- Hardware Design and Testing
    - Data from bench/lab testing of power shuffler version two and MPPT version one
    - Updated PCB design for power shuffler version three and MPPT version two
    - Data from lab/field testing of power shuffler version three and MPPT version two
    - Optional/ Unknown
        - Additional hardware revision as necessary during buffer period
- Documentation
    - Project Report
        - Optional Draft
    - Handover documentation
    - Presentation Slides

Risks, Requirements & Constraints

Requirements

The power shuffler hardware must satisfy the following functional and performance requirements: The device must support bidirectional power flow between adjacent PV modules within a string, as this is fundamental to the current redistribution principle of the shuffler topology. Overload protection must be implemented to prevent damage to the hardware and connected PV modules under fault conditions. The shuffler must be rated to handle the output of the highest commercially available PV module at time of design; the Aiko solar half-cut cell panel, rated at approximately 500W, has been selected as the reference module to ensure broad compatibility with current and near-future PV products. A minimum switching frequency of 10kHz is required to ensure adequate ripple suppression and to keep passive component sizes manageable.

The design must account for the following partial shading scenarios as validation test cases: single whole-panel shading, half-panel shading, single-string shading, multiple modules within a string shaded, and single module shading. These scenarios represent the range of real-world conditions the shuffler is expected to mitigate.

The input to the shuffler is defined by the highest-rated commercially available PV panel to ensure forward compatibility. The output interface assumes integration with an MPPT boost converter, with loads limited to energy storage systems (ESS) only for the scope of this project.

Constraints

The project is subject to several constraints. From an electrical safety standpoint, all hardware and testing must comply with AS3000 Australian Wiring Rules. As the system may operate above 150V DC when connected to a PV string, a licensed electrical technician must be present and supervising

during any live system testing. This requirement directly influences the testing timeline and access to QUT's Power Engineering Research Facility.

The project is time-constrained to under one year across two semesters and a summer break period, and is limited to a team of two students working across the power electronics and control systems streams respectively. Component selection is constrained to commercially available parts to ensure reproducibility, reasonable lead times, and cost control. Budget constraints have not been formally defined at this stage and will require confirmation with the supervisor.

There are existing commercial products in the power optimiser space, most notably those produced by SolarEdge, which employ similar per-module power optimisation concepts and hold associated intellectual property. As this project is conducted solely for academic research purposes and not for commercial deployment, IP conflict risk is considered low. However, care should be taken during documentation and any future publication to ensure the novel aspects of this design are clearly distinguished from existing patented implementations.

Risks

The primary risks identified for this project are as follows: Scope creep represents a significant risk given the breadth of the design, which contains simulation, multiple PCB revisions, and field testing across two hardware streams. This will be mitigated by strict adherence to the staged deliverable schedule and regular supervisor check-ins to rebaseline scope if required. Electrical safety risk is present throughout bench and field testing phases due to the elevated voltages involved in live PV string testing; this is mitigated by conducting high-voltage testing exclusively at QUT's Power Engineering Research Facility under supervision of a licensed technician, and by following an incremental risk protocol starting from low-power bench emulation before progressing to live panels. Component lead time risk may impact PCB revision turnaround, particularly over the summer break period; this is mitigated by early ordering and maintaining a list of pre-approved alternative components for critical parts. Finally, design interdependency risk exists between the power electronics and control systems streams, a delay or design change in one stream can impact the other. This is managed through clearly defined interface specifications agreed upon early in the project.

Quality & Sustainability

The quality of each stage of the project will be assessed by the supervisor and project member due to the reliance of the controls and power electronics components on each other. The quality of circuit simulations will be validated by comparing simulation outputs against theoretical calculations derived from the PV single diode model and known converter characteristics. Hardware quality will be assessed through a structured bench testing protocol, where key performance metrics such as switching efficiency, voltage/current ripple, and thermal behaviour are measured and compared against simulation results and design targets. Discrepancies beyond an acceptable margin will trigger a design review prior to progressing to the next hardware revision.

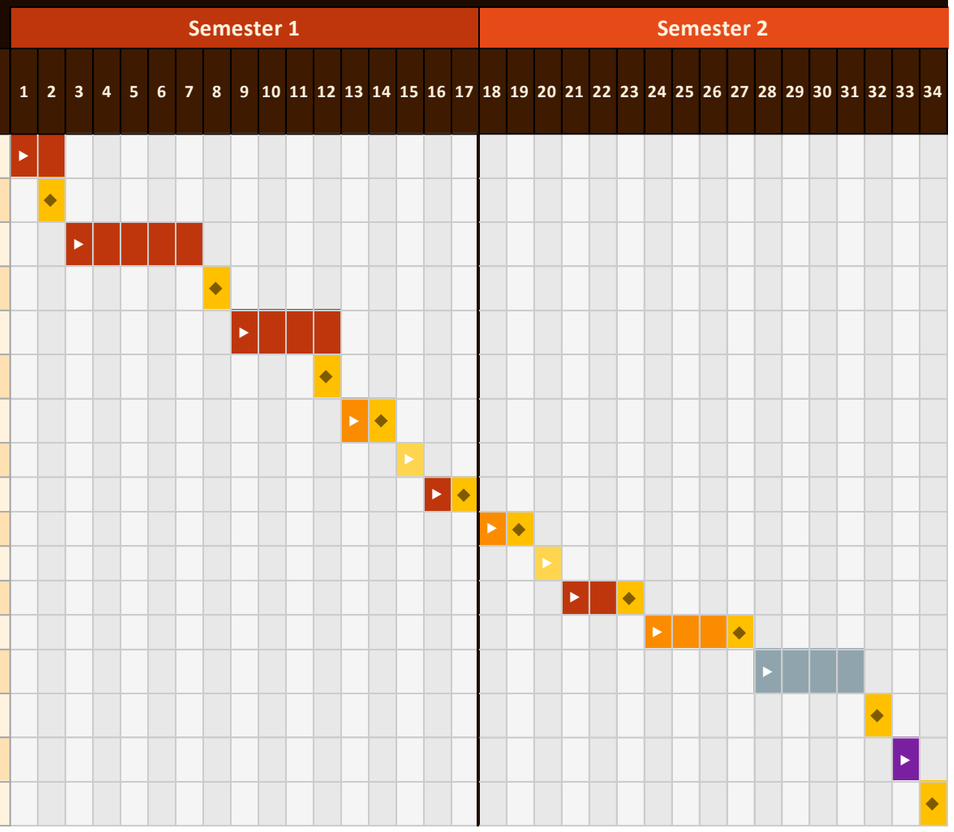
From a sustainability perspective, the hardware is designed with longevity and repairability as primary considerations rather than single-use deployment. Component selection will prioritise industrial-grade or automotive-grade rated parts, particularly for MOSFETs, capacitors, and gate drivers, which offer extended operational lifespans under thermal and electrical stress compared to consumer-grade equivalents. The modular PCB architecture across multiple revisions is intentional in this regard; rather than discarding entire assemblies when issues are identified, individual boards

can be repaired or upgraded independently, reducing material waste throughout the development lifecycle. Connectors and interfaces are selected to be standardised where possible, ensuring long-term replaceability without requiring full system redesign. The use of a universal power shuffler design compatible with a wide range of existing PV modules further extends the useful life of the hardware by avoiding obsolescence tied to any specific panel generation. Long-term sustainability at the system level is also considered by improving energy extraction efficiency from existing PV installations through partial shading mitigation, the hardware reduces the need for additional panel installations to meet equivalent energy output, thereby lowering the embodied energy and material cost associated with expanding a solar array.

Timeline & Deliverables

Power Electronics Sub-Timeline | Solar Power Shuffler Hardware

#	Phase	Task / Deliverable	Team	Dependencies	Start Wk	End Wk	MS
1	Setup	Project Setup	Both	—	1	2	
2	Setup	1st Supervisor Meeting & Communication Strategy	Both	T01	2	2	M1
3	Planning	Project Planning & Literature Review	Both	T02	3	7	
4	Design	High-Level System Architecture	Both	T04	8	8	M2
5	Design	Shuffler V1: Power Circuit & Hardware Design	Power Elec.	T05	9	12	
6	Design	Hardware Simulation & BOM Documentation	Power Elec.	T22	12	12	M3
7	Testing	Hardware Unit Testing – Shuffler V1	Power Elec.	T22, T07	13	14	M4
8	**Order**	**PCB V2 & MPPT Board Ordered**	**Power Elec.**	**T23, T24**	**15**	**15**	
9	Design	PCB Design: Pair Configuration & MPPT	Power Elec.	T25	16	17	M5
10	Testing	PCB Subsystem Assembly & Verification	Power Elec.	T25, T26	18	19	M6
11	**Order**	**PCB V3 & MPPT V2 Ordered**	**Power Elec.**	**T27**	**20**	**20**	
12	Design	PCB String Configuration Design	Power Elec.	T28	21	23	M7
13	Testing	Full-System Hardware Validation	Power Elec.	T28, T29	24	27	M8
14	Buffer	Buffer / Contingency	Both	—	28	31	
15	Close	Final Report & Handover Documentation	Both	T21	32	32	M9
16	Close	Presentation Preparation	Both	T32	33	33	
17	Close	Presentation Delivery	Both	T33	34	34	M10



Management of Project Changes

Given the iterative and hardware-intensive nature of this project, changes to scope, design, or timeline are considered likely and are anticipated as a normal part of the development process rather than an exception. Each PCB revision cycle inherently carries the possibility of uncovering issues that necessitate design changes, and real-world testing with PV modules may reveal performance characteristics that diverge from simulation, requiring adjustments to the shuffler topology, component selection, or control strategy.

Any proposed change to the project scope, deliverables, or timeline must first be discussed with and approved by the supervising academic before implementation. Once approved, this Scope of Work document will be updated to reflect the change, with the version table on the cover page updated accordingly to record the nature of the change, the date, and the author. This ensures a clear audit trail of how the project has evolved over time.

For changes arising from hardware testing outcomes such as the need for an additional PCB revision or a shift in topology a brief written summary of the issue identified, the proposed change, and its impact on downstream deliverables will be prepared and presented to the supervisor at the next scheduled meeting. Buffer weeks have been deliberately allocated in the Semester 2 timeline to absorb changes of this nature without requiring a full reschedule of the project.

Changes that affect the interface between the power electronics and control systems streams must be communicated between both students promptly, with updated interface specifications agreed upon and documented before either stream proceeds. This is particularly important for communication protocol decisions and microcontroller integration points, where an unannounced change by one stream could invalidate work already completed by the other.

Sign off

	SIGNATURE	DATE
STUDENT ENGINEER	Son Nguyen	15 April 2026

Appendix: Review of Literature

In the last 10 years, photovoltaic cells usage in the role of solar energy generation have become increasingly common due to improvements in collection efficiency and reduction in price. These two factors have led to a massive increase in their utilisation on a global scale. Between 2014 and 2024, annual PV installations have increased 15-fold from 40GW to 602GW of annual PV installations. From an Australian perspective, the annual solar electricity generation revenue has increased from $100 Million AUD in 2015 to $2.1 Billion AUD in 2025.

Despite improvements being made to PV cells and modules, their connections have remained roughly the same ever since their inception. On a typical residential and commercial PV panel, cells are arranged in series in a grid pattern. Depending on the number of cells on the panel, bypass diodes are placed in parallel between every 20 or 24 cells for a panel with 60 or 72 cells respectively (creating 3 sub-strings) to prevent damages to the panel in the instances of partial shading. Several panels are then placed in series (forming a string) to increase the output voltage to the desired output bus level, where a DC – AC inverter operates and export the power to the grid. Several strings are then placed in parallel to increase the output power from an array whilst limiting voltage to reasonable levels.

To understand the purpose of bypass diodes, the single diode diagram of PV cells/module will now be discussed as seen in figure 1 below.

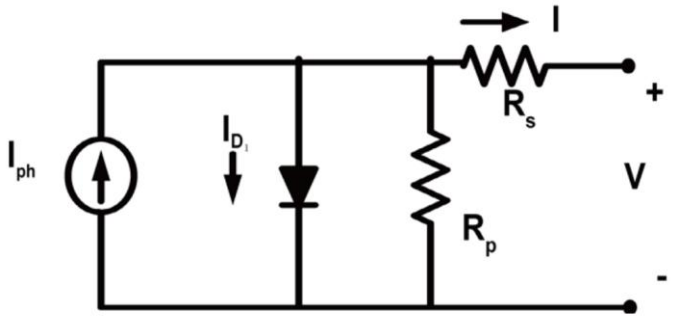


Figure 1 - PV Single Diode Diagram

The output characteristics of a PV cell using this model could be summarised using the following five equations

$$I = I_{ph} - I_o\left[\exp\left(\frac{V + IR_s}{aV_T}\right) - 1\right] - \frac{V + IR_s}{R_P} \quad (1)$$

$$V_t = \frac{N_s kT}{q} \quad (2)$$

$$I_{ph} = \frac{G}{G_n}\left[I_{pvn} + K_1(T - Tn)\right] \quad (3)$$

$$I_o = I_{on}\left(\frac{Tn}{T}\right)^3 \exp\left[\frac{qE_g}{ak}\left(\frac{1}{T_n} - \frac{1}{T}\right)\right] \quad (4)$$

$$I_{on} = \frac{I_{scn}}{\exp\left(\frac{V_{0cn}}{\alpha V_{Tn}}\right) - 1}$$

Where:

I_{ph}: Photocurrent (Generated by the PV when lights hits the semiconductor material)

I_{pvn}: Nominal light current (current generated at test conditions)

I_o: Saturation Current

I_{on}: Nominal Saturation Current

I_{scn}: Nominal Short Circuit Current

G: Operating Irradiance (in W/m^2)

G_n: Nominal Irradiance (reference sunlight level, typically 1000W/m^2)

K_1: Temperature coefficient

V_T: Thermal Voltage

V_{ocn}: Nominal Open Circuit Voltage

N_s: Number of cells in series

k: Boltzmann constant

T: Cell temperature

T_n: Nominal temperature (typically 25 degrees Celsius)

q: Electron Charge

E_g: Bandgap enery

α: Ideality Factor

As seen in equation (3) and (1). A reduction in Irradiance (G) will reduce in lower photocurrent generation (I_{ph}). Thus, the total output current (I) will be reduced after accounting for losses. Thus, when a shaded cell/modules is placed in series with other unshaded cells/modules, two possibilities can occur. In the first instance, the entire output current of the string will be reduced to match that off the partially shaded cell/module. In the more extreme instance of complete shading ($G = 0$) of cells/ module, I_{ph} reduces to 0 and the PV cell/module will become reverse biased and act as a current sink ($I < 0$). In this instance, the cell begins to act as a load within the panel and create a hotspot due to its current draw. It should be noticed that PV cells become reversed biased when ($I < 0$) and does not require complete shading of the cell. To prevent thermal degradation and flammable risk to the PV cell generated by the hotspots, bypass diodes are placed in parallel between several cell to provide an alternative current flow path once a cell/cells become reverse biased.

Whilst bypass diodes fulfil the objective of protecting PV cells from thermal damage, they can excessively reduce the output power of a given module as the current will be bypassed around the entire substring through the bypass diode. On a modern PV panel with 72 cells and conventional bypass diode layout (one diode placed in parallel with 24 cells), a one-third reduction of the panel's maximum output power can occur due to a single cell being shaded.

Due to the effect of irradiance (G) and losses associated with solar generation from PV cells as seen in equation (1), the output of PV modules are nonlinear. As such, PV's modules output are highly dependent on the connect load and an appropriate load should be set so that the maximum amount of power could be extracted from them. The load point at which the maximum power could be extracted is known as the maximum power point (MPP) as seen in Figure 2 below.

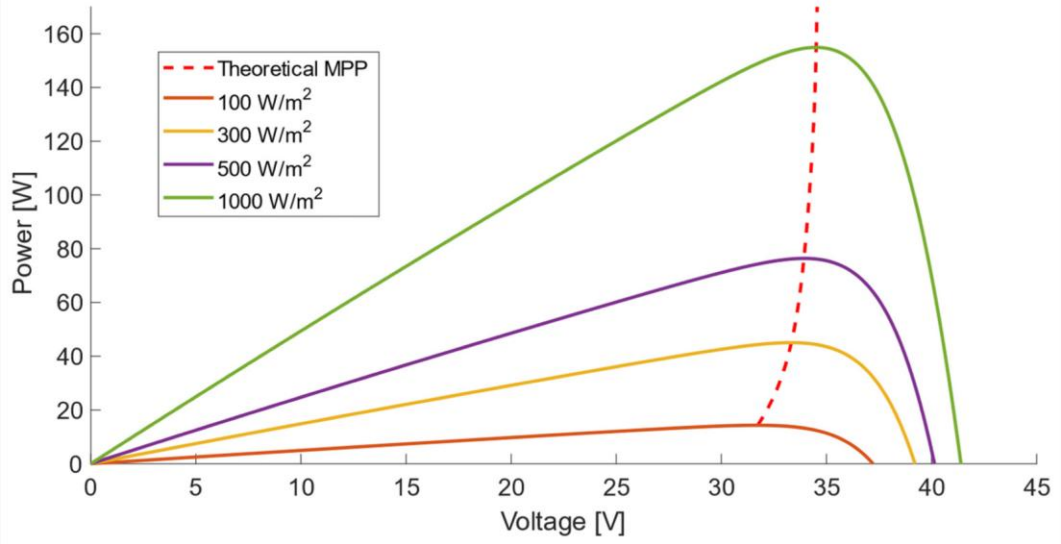


Figure 2 - Output Power vs Voltage Curve and theoretical MPP of UP-M155M panel

Maximum Power Point Tracking (MPPT) is an electronic control system designed to continuously vary the electrical load of a PV module, ensuring it operates at its Maximum Power Point (MPP) regardless of atmospheric conditions. Given the non-linear output characteristics defined by the current-voltage relationship in equations (1) through (5), the optimal operating point is highly sensitive to external disturbances such as shifting irradiance and temperature. Therefore, the implementation of an MPPT controller is required to dynamically track these fluctuations in real-time; forgoing such a system would force the PV module to operate at a fixed, sub-optimal point on the P-V curve, leading to significantly lower energy extraction efficiency as illustrated in Figure 2.

MPPTs are the most effective when the PV modules are operating in unshaded or uniformly shaded conditions as the output characteristic of the connected PV system will present only one global MPP maxima as seen in Figure 2. However, in an instance where two PV panels are connected in series with one being shaded, each panel will be operating at different output curves. As they are connected in series, the output curve of the string will be summation of the two panels in series and presents two local maxima as seen in Figure 3.

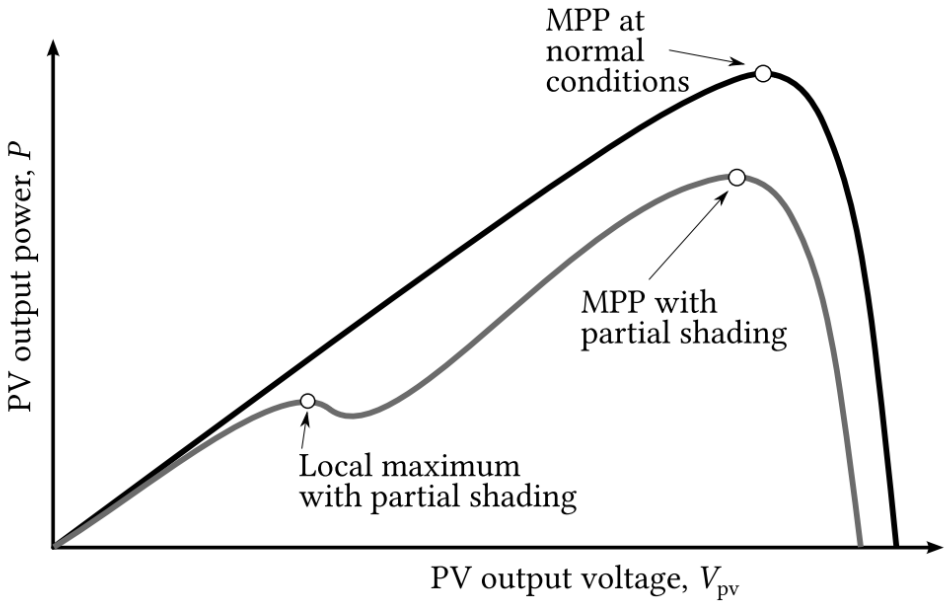


Figure 3 - Power Curve of a PV Module/String with partial shading

Due to their low complexity, computation requirements and minimum cost, the Perturb and Observe algorithm (P&O) have grown to become the most common method for MPPT, especially in solar applications. However, their simple logic makes them susceptible to oscillating around a power maxima point. Whilst this behaviour is desired where only one global maxima exists (such as non-shading or uniform shading condition), it can cause the MPPT to oscillate around an undesirable local maximum with partially shaded module/strings as seen in figure 3, leading to reduced power extraction. In large arrays with panels receiving different irradiance (shading), multiple local maximums exist, which further exacerbate the problem of non-optimal tracking by the P&O algorithm.

Within the last decade, interests in PV arrays have become increasingly popular due to their growing presence in the energy sector with many targeting the problem of partial shading. Within these research, two primary approaches have been suggested to alleviate or solve the partial shading problem, namely: Array reconfiguration, control systems algorithms and power shufflers devices.

Array reconfiguration consists of dynamic array reconfiguration (DAR) and static array reconfiguration (SAR) and refers to PV systems where each module in a string are electrically connected via switches (e.g. MOSFETs) to the equivalent module in the parallel string (e.g. all modules in row 1 are connected). In DAR systems, modules are rearranged in real-time based on changing conditions so that each module in a string receives the same amount of irradiance or each row in a system receives the same amount irradiance (e.g. cell 1 in string 1 receivers the same amount of irradiance as cell 1 in string 2, etc.). DAR system aims to either prevent output current mismatch or ensures that the output characteristic of each string remains the same, preventing multiple local maximum power points from occurring. In SAR systems, the electrical connection of each module are arranged in a fixed during before installation and modules in each row are not electrically connected. SAR's goal is to disperse shading across the array as evenly as possible to ensure similar output characteristic between strings. Whilst this technology can provide superior energy collection efficiency, its wiring complexity and costs (especially for DAR) has limited their real-world uses.

Research into power shuffler devices have become increasingly popular over the last two decade as the effects of partial shading becomes more well understood. As per the name, these devices work on the principle of distributing a string's generated current so that every module within the string has the same output current. Whilst power shufflers represents one approach to the partial shading problem, they sits within a broader family of panel-integrated DC-DC converters which have been proposed in the literature, each with their own trade-offs between efficiency, complexity and cost. To understand where the power shuffler fits within this design space, the available converter topologies will now be discussed.

Kasper et al. [X] provides a comprehensive classification of PV panel-integrated DC-DC converter topologies suitable for partial shading mitigation. These converters can be broadly categorised into two groups based on the proportion of panel power being processed: full-power converters and partial-power converters

Full-power converters process the entire output of a PV panel before it is fed into the string and can be further divided into series and parallel connected subcategories. Within the series connected subcategory, three topologies are commonly used, namely the buck, boost and buck-boost converters. As per their names, buck converters compensates for shading by stepping down the voltage of shaded panels, increasing their output current to match that of the unshaded panels. Boost converters operate in the opposite manner by stepping up the voltage of unshaded panels to

reduce their output current to match that of the shaded panels. Buck-boost converters offers the greatest flexibility as they can perform both operations, allowing any string current to be matched regardless of the shading condition [X]. However, as the entire panel output must pass through the converter, conversion losses are incurred even in unshaded conditions where no mismatch compensation is required. Parallel connected full-power converters such as high step-up boost and flyback converters are also possible but are less common in series string applications due to their higher complexity.

Partial-power converters (PPCs) process only a fraction of the panel's output power, with the remainder of the power flowing directly from the panel to the string output through a highly efficient bypass path. Due to this arrangement, PPCs provides an inherent efficiency advantage over full-power converters as the effective conversion efficiency approaches 100% in the ideal case where no mismatch is present [X]. PPCs are further categorised into series-connected partial-power converters (S-PPCs) and parallel-connected partial-power converters (P-PPCs). S-PPCs compensate for mismatch by adding or subtracting a voltage in series with the panel output and are typically realised using a full-bridge or push-pull stage on the primary side with a diode rectifier on the secondary side of an isolation transformer. The P-PPC concept, also known in the literature as the "energy shuffler" or "current diverter" [X], operates on the principle of voltage equalisation between adjacent panels. By connecting a bidirectional converter between neighbouring modules, excess current from unshaded panels is diverted to shaded panels, allowing all panels in the string to operate close to their individual MPP voltage despite the differences in their MPP currents. Common P-PPC realisations as seen in the literature includes the bidirectional buck-boost converter, the Ćuk converter and the flyback converter.

For this project, the P-PPC architecture has been selected as the basis for the power shuffler design, building on the foundational work by Walker [Y]. This architecture was chosen due to the inherent efficiency advantage of partial-power processing as well as the fractional power rating that this approach allows, both of which are well suited to a scalable and cost effective design. The specific topology is yet to be finalised, with the non-isolated bidirectional buck-boost and Ćuk converters currently being considered as the most likely candidates. Both topologies are capable of bidirectional current shuffling between adjacent modules without the need for an isolation transformer, reducing component count and cost in comparison to isolated alternatives such as the flyback converter. The final selection will be made during the simulation phase of the project based on factors such as component stress, ripple current, efficiency and ease of integration with commercially available PV modules.

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